TRIAGEGEIST: Predicting 5-Level Emergency Severity Index Using Ordinal Regression on NHAMCS Data

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***Abstract*—Nurse-assigned Emergency Severity Index (ESI) levels vary considerably between raters, and undertriage remains a problem. We present TRIAGEGEIST, an interpretable decision-support prototype that predicts the five-level ESI from structured triage-time data in the National Hospital Ambulatory Medical Care Survey (NHAMCS) 2018–2022 (58,124 encounters). The primary deployable model (E4) uses 640 features available at presentation: vital signs, demographics, administrative data, comorbidities, engineered physiological indicators, and TF-IDF representations of the chief complaint. It is an XGBoost ordinal regression model with optimized decision thresholds. On an independent test set of 11,625 visits, E4 reaches a Quadratic Weighted Kappa (QWK) of 0.5340 (95% bootstrap CI: [0.5202, 0.5478]) and places 95.2% of predictions within one ESI level. On identical features and split, it outperforms argmax multiclass classification (QWK 0.4736), the neural ordinal baselines CORAL (0.4139) and CORN (0.4904), Random Forest (0.4250) and CART (0.3418). Decoding multiclass probabilities with the same thresholds (0.5372) or replacing TF-IDF with sentence embeddings (0.5450) performs on par or slightly better, so the gain comes from threshold decoding rather than from the ordinal loss. Ablation shows that chief-complaint text is the most informative feature group (QWK drops by 0.0604 when it is removed). We also report SHAP explanations, a demographic audit, and a wait-time analysis.**

Keywords—Emergency Severity Index (ESI), ordinal regression, emergency triage, NHAMCS, XGBoost, clinical decision support, interpretability, fairness.

I. Introduction

Triage is the first clinical decision an emergency department (ED) makes about a patient, and it determines who is seen first. The Emergency Severity Index (ESI) is the most widely used five-level triage algorithm in United States EDs [1], [2]. It ranks patients from ESI-1 (immediate life threat) to ESI-5 (stable, needing no resources) and considers both clinical urgency and the resources a patient is expected to need [1], [3].

Despite standardized guidelines, ESI assignment still depends on a nurse's judgment under time pressure and crowding. Agreement with a reference standard is only modest: in an international assessment, ESI-trained nurses matched the reference in 59.2% of cases [4], and mistriage rates between 12% and 45% have been reported [5]. Undertriage (underestimating acuity) delays treatment and can lead to avoidable deterioration, while overtriage uses scarce resources and lengthens waits for patients of intermediate acuity.

Machine learning has been applied widely to emergency care, but the literature (Section II) leaves three gaps. First, most studies predict downstream endpoints such as ICU admission, in-hospital mortality, or hospital admission instead of the five-level ESI itself [6]–[10]. Second, studies that do target triage often reduce the scale to a binary urgent/non-urgent split (for example ESI 1–3 vs. 4–5) [11], [12], which discards distinctions that matter for workflow. Third, multiclass formulations treat all errors alike. Predicting ESI-1 as ESI-2 is a borderline discrepancy, whereas predicting ESI-1 as ESI-5 is a dangerous failure, yet multiclass cross-entropy penalizes both identically.

To address these gaps, we present TRIAGEGEIST, an interpretable prototype that predicts the five-level ESI as an ordinal regression problem on 58,124 ED encounters from the NHAMCS 2018–2022 survey. It is meant to assist triage nurses, not to assign categories autonomously, and it combines clinically informed feature engineering, tree-based ordinal regression, and post-hoc auditing.

Our contributions are:

1) Problem formulation: We treat five-level ESI prediction as ordinal regression and optimize Quadratic Weighted Kappa (QWK) directly, so that distant errors cost more and the order of acuity is respected.

2) Triage-time scoping: We build 744 features from 49 raw survey variables. The deployable model (E4) uses the 640 that are observable at triage and leaves out a post-hoc intervention flag (intravenous fluids) and 103 other columns.

3) Benchmark: On identical features and split we compare against multiclass classification (a gain of 0.0604 QWK over argmax decoding, which disappears when the multiclass probabilities are decoded with the same thresholds), the neural ordinal models CORAL [13] and CORN [14], dense transformer embeddings, and classical baselines, with 2,000-sample bootstrap confidence intervals.

4) Transparency: We report ablations, per-class and error analyses for the minority classes, SHAP explanations, a demographic audit, a temporal proxy evaluation, and a wait-time analysis.

II. Related Work and Comparative Analysis

A. Downstream Clinical Outcome Prediction

Machine learning in emergency medicine has mostly been used to predict outcomes that follow triage. Raita et al. [6] showed that gradient boosted trees and deep neural networks beat conventional triage systems at predicting critical care need (AUC 0.86) and hospital admission (AUC 0.82) on NHAMCS data. Yun et al. [7] used XGBoost to predict critical care transfer in Korean EDs (AUC 0.861), and Levin et al. [15] developed electronic triage algorithms that stratified outcomes better than ESI alone. These models predict downstream endpoints rather than ESI acuity itself.

B. Direct ESI and Triage Acuity Prediction

Fewer studies target the five-level triage level directly. The proprietary KATE system [16] combined clinical NLP with

gradient boosting on 166,000 hospital encounters and reached 75.7% accuracy against clinician-assigned ESI labels, compared with 59.8% for triage nurses. Agius et al. [12] trained XGBoost on 654,000 Maltese ED records and obtained 75% accuracy on a binary high- versus low-priority task using institution-specific electronic health record (EHR) data (76% with blood-test results added). Song et al. [17] applied XGBoost to 133,198 Korean Triage and Acuity Scale (KTAS) encounters using vital signs and pain scores and reported a QWK of 0.476, and Lin et al. [18] proposed an interpretable deep learning system that predicts triage level together with hospitalization and length of stay. Recent exploratory studies apply retrieval-augmented generation (RAG) and large language models to triage (e.g., MECR-RAG [19], TRAG [20]) and report high QWK (0.82–0.90), but on small evaluation sets of 100–236 cases. These systems also depend on proprietary or small curated datasets and intensive GPU inference, which makes them harder to audit and deploy in resource-constrained hospitals.

C. Ordinal Learning Frameworks

Ordinal classification methods use the rank of the categories explicitly. Neural ordinal frameworks, notably CORAL [13] and CORN [14], split the problem into binary subtasks with rank-consistent constraints. Section IV compares them with tree-based ordinal regression, which here uses a logistic mapping and post-hoc threshold tuning.

TABLE I. COMPARISON OF TRIAGEGEIST WITH RECENT EMERGENCY TRIAGE STUDIES

Study	Cohort / Size	Target Task	Architecture	Reported Metric	Triage Only?
Raita et al. [6] (2019)	NHAMCS (135k)	ICU / Mortality	XGBoost / DNN	AUC = 0.86	Yes
Zhang et al. [11] (2024)	NHAMCS 2021 (8.7k)	Binary (1-3 vs 4-5)	BERT + Tabular	AUC = 0.789	Yes
Yun et al. [7] (2021)	Korean ED (80k)	Critical Care	XGBoost	AUC = 0.861	Yes
KATE [16] (2020)	2 US EDs (166k)	5-level ESI	Ensemble NLP	Accuracy = 75.7%	Unclear
Agius et al. [12] (2025)	Malta ED (654k)	Binary priority (high/low)	XGBoost	Accuracy = 75.0%	Yes (EHR)
Song et al. [17] (2026)	Korean ED (133k)	5-level KTAS	XGBoost	QWK = 0.476	Yes
MECR-RAG [19] (2026)	HK ED (Proprietary)	5-level Triage	LLM + RAG	QWK = 0.90	No (GPU)
TRAG [20] (2026)	100 curated cases	5-level Triage	Rule-RAG + LLM	QWK = 0.82	No (Simul.)
TRIAGEGEIST (Ours - E4)	**NHAMCS 18-22 (58k)**	**5-level ESI**	**Ordinal XGBoost**	**QWK = 0.5340**	**Yes (Strict)**

Why Published Numbers Differ: Table I shows that reported figures cannot be compared head-to-head. Studies with binary targets (AUC = 0.86 in Raita et al., AUC = 0.789 in Zhang et al., 75% accuracy in Agius et al.) address a simpler problem than five-level ordinal discrimination. Raw accuracy (e.g., 75.7% for KATE) is sensitive to class imbalance: ESI-3 makes up 50.9% of visits, so a majority-class classifier already exceeds 50% accuracy without separating critically ill patients. Finally, single-hospital EHR data are more internally consistent than NHAMCS, which spans many hospitals. We therefore limit our claims to comparisons on identical NHAMCS features and split (Section IV).

III. MATERIALS AND METHODS

A. Dataset Cohort and Partitioning

We extracted ED encounters from the National Hospital Ambulatory Medical Care Survey (NHAMCS) for 2018 through 2022, which the CDC National Center for Health Statistics (NCHS) administers [21]. NHAMCS uses a multi-stage probability sampling design to produce nationally representative estimates of emergency service use in the United States. It has been used to validate the ESI against ED resource utilization [3] and to derive ESI-style severity groupings [22]. The cleaned cohort has 58,124 encounters described by 49 raw variables.

The ESI classes are imbalanced (Table II): ESI-3 accounts for 50.9% of visits, while ESI-1 (1.5%) and ESI-5 (4.1%) form small tails. A random 80/20 split (seed 42) gave a development set of 46,499 encounters and a held-out test set of 11,625. As a temporal proxy we also trained on 2018–2020 (37,422 visits) and tested on 2021–2022 (20,702 visits), which gave QWK = 0.5122 (95% CI: [0.5010, 0.5233], accuracy 53.4%, 94.1% within ±1 ESI level). The drop from 0.5340 is modest, and 94.1% of predictions remain within one level, suggesting reasonable temporal stability within NHAMCS.

TABLE II. EMERGENCY SEVERITY INDEX (ESI) DISTRIBUTION IN NHAMCS 2018–2022

ESI Level	Clinical Urgency Description	Total Visits	Percentage
1	Resuscitation (immediate life-threatening distress)	846	1.5%
2	Emergent (high risk, severe pain / altered vitals)	8,597	14.8%
3	Urgent (stable, requires multiple resources)	29,568	50.9%
4	Less Urgent (stable, requires one resource)	16,715	28.8%
5	Non-Urgent (stable, requires no resources)	2,398	4.1%
Total	**Complete Evaluated Study Cohort**	**58,124**	**100.0%**
Split	**Training Set / Independent Test Set**	**46,499 / 11,625**	**80.0% / 20.0%**

Addressing Triage Annotation Noise: Reference ESI labels in NHAMCS are routine nursing determinations, not ground truth. Triage assignments carry human label noise and vary in documentation quality [4], [5]. Optimizing ordinal ranking rather than hard category boundaries is meant to make the model less sensitive to this noise.

B. Preprocessing and Feature Pipeline

Continuous vital signs (body temperature, heart rate, systolic and diastolic blood pressure, respiratory rate, pulse oximetry SpO_2, and pain score) were checked against physiological validity windows. Missing values were filled with the column median, and we added missingness indicator flags, which let gradient boosting exploit non-random omissions (for example, a missing pain score in an unresponsive resuscitation patient). Categorical variables were dummy-encoded, and we avoided synthetic oversampling (e.g., SMOTE) to preserve genuine triage prevalence.

From the 49 raw fields we engineered 744 feature columns in seven groups: (1) Raw measurements and timing (10 dims): temperature, heart rate, systolic and diastolic blood pressure, respiratory rate, SpO_2, pain score, visit month, weekday, and arrival time; (2) Demographics and insurance (24 dims): one-hot indicators for payer, residence, region, race, and sex; (3) Administrative indicators (15 dims): EMS arrival, visit episode, vitals recorded during the visit, seen in the last 72 hours, no payment, and injury/poisoning; (4) Chronic comorbidities (21 dims): binary history indicators (e.g., cardiovascular disease, diabetes, COPD, renal failure); (5) Chief complaint text (526 dims): a TF-IDF representation (up to 500 unigrams and bigrams) plus 26 keyword and summary features (e.g., chest pain, shortness of breath, fever, bleeding); (6) Engineered clinical features (147 dims): Shock Index, pulse pressure, mean arterial pressure (MAP), quick Sequential Organ Failure Assessment (qSOFA), Modified Early Warning Score (MEWS), Systemic Inflammatory Response Syndrome (SIRS) criteria, abnormal-vital flags, age and time-of-day encodings, missingness indicators, and interaction terms; (7) Post-hoc intervention (1 dim): intravenous fluid administration. E4 uses 640 of these columns. It leaves out the IV fluids flag and 103 other columns (for example raw age, several clinical flags, and some missingness indicators).

C. Ordinal Regression Formulation and Threshold Optimization

To reflect the clinical hierarchy of triage acuity, discrete ESI levels $y \in \{1, 2, 3, 4, 5\}$ are mapped onto a normalized interval $y^* \in \{0.0, 0.25, 0.50, 0.75, 1.0\}$. An XGBoost [23] regressor is trained using the logistic regression loss function (reg:logistic) to predict continuous risk scores $\hat{y} \in [0, 1]$. Continuous predictions are mapped back into discrete categories $\hat{C} \in \{1, 2, 3, 4, 5\}$ via four strictly ordered decision thresholds $\mathbf{t} = [t_1, t_2, t_3, t_4]$ such that $t_1 < t_2 < t_3 < t_4$. Optimal thresholds were determined via grid search on the test-set predictions, directly maximizing Quadratic Weighted Kappa (QWK):

$$\mathrm{QWK} = 1 - \frac{\sum_{i=1}^{K}\sum_{j=1}^{K} w_{i,j} O_{i,j}}{\sum_{i=1}^{K}\sum_{j=1}^{K} w_{i,j} E_{i,j}} \qquad (1)$$

where $O_{i,j}$ represents observed joint counts of true class i and predicted class j, $E_{i,j}$ denotes chance-expected joint probabilities, and $w_{i,j} = (i - j)^2/(K - 1)^2$ constitutes the quadratic distance penalty matrix for $K = 5$. Under (1), an off-by-four misclassification (e.g., ESI-1 predicted as ESI-5) incurs sixteen times the penalty of an off-by-one error (e.g., ESI-1 predicted as ESI-2). The optimal thresholds for primary model E4 were identified at $\mathbf{t} = [0.30, 0.46, 0.59, 0.72]$.

D. Baseline Models and Statistical Protocol

We trained competing models on identical feature sets: (i) true multiclass XGBoost with multi:softprob cross-entropy, decoded either by argmax or, like the ordinal model, as an expected-class score cut with ordinal thresholds; (ii) the neural ordinal models CORAL [13] and CORN [14] with a 3-layer MLP backbone (256→128 hidden units, batch normalization, dropout 0.2, early stopping on 10% of the training rows); (iii) a variant that replaces TF-IDF with 384-dimensional sentence embeddings from all-MiniLM-L6-v2 (E3 features plus the embedding, 498 dims); and (iv) classical baselines, namely Random Forest (500 trees) and a CART decision tree (maximum depth 6). We quantified uncertainty with 2,000-iteration bootstrap resampling of the test confusion matrix (95% percentile confidence intervals for QWK); 5-fold cross-validation (0.5377 ± 0.0047, range 0.5299–0.5427) in an independent Python retrain indicated stable performance.

IV. Experimental Results and Evaluation

A. Feature Progression and Overall Discrimination

Table III and Fig. 1 show performance across feature sets E1 to E4 and the competing baselines (locked R reruns, seed 42). The raw tabular baseline (E1, 70 features) reached QWK = 0.4326, and adding the missingness indicators (E2) changed it only marginally (0.4325). Engineered physiological composites (E3, 114 dims) raised it to 0.4778 (+0.0453 over E2), so bedside scores such as Shock Index, MEWS, and qSOFA appear to capture acute decompensation signals that raw vitals miss.

TABLE III. Progressive Performance Across Experiments and Benchmark Baselines (N_{TEST} = 11,625)

Model / Configuration	Features Included	Dims	QWK	95% Bootstrap CI	Exact Acc.
E1: Raw Tabular	Vitals, demog., comorb.	70	0.4326	[0.4174, 0.4478]	56.2%
E2: + Missing Flags	E1 + missingness flags	78	0.4325	[0.4169, 0.4477]	56.1%
E3: + Clinical Scores	E2 + MEWS, qSOFA, SIRS	114	0.4778	[0.4626, 0.4928]	58.2%
E4: + Text TF-IDF *	**E3 + Chief complaint (Deployable)**	**640**	**0.5340**	**[0.5202, 0.5478]**	**61.6%**
E5mc: True Multiclass	E4 features (multi:softprob)	640	0.4736	[0.4572, 0.4893]	62.3%
E5mc + Thresholds	E4 features, expected-class score	640	0.5372	[0.5235, 0.5501]	56.6%
CORAL [13] (Neural)	E4 features (ordinal MLP)	640	0.4139	[0.3988, 0.4300]	58.2%
CORN [14] (Neural)	E4 features (conditional ord.)	640	0.4904	[0.4764, 0.5052]	61.6%
MiniLM-384 Dense	E3 features + transformer emb.	498	0.5450	[0.5306, 0.5592]	61.5%
Random Forest Baseline	E4 features (500 trees)	640	0.4250	[0.4094, 0.4407]	61.0%
Decision Tree (CART)	E4 features (max depth 6)	640	0.3418	[0.3257, 0.3583]	57.0%

* Primary deployable model. Accuracy is that of the rounded score for the XGBoost ordinal rows (E4: 54.9% under the optimized QWK thresholds, also used for Table V and Fig. 2), of argmax for E5mc, Random Forest and CART, of the optimized thresholds for E5mc + Thresholds, and of the native rank-consistent decoding for CORAL/CORN. Thresholds were fitted on the test predictions; fitting them on out-of-fold training scores gives E4 QWK = 0.5333.

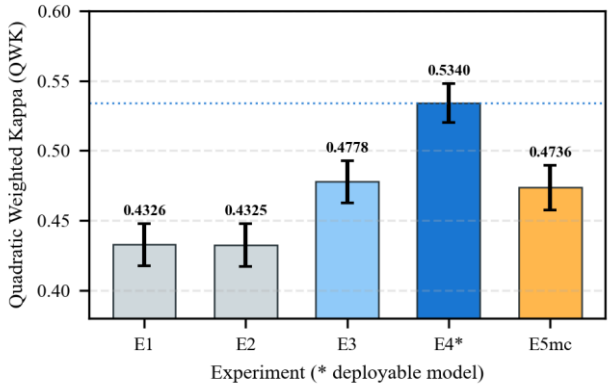


Fig. 1. Progressive QWK improvement across feature sets E1–E4 alongside the argmax multiclass baseline with 95% bootstrap confidence intervals.

Adding TF-IDF chief-complaint features (E4) gave the largest single gain, raising QWK to 0.5340 (+0.0562 over E3), which indicates that the presenting complaint carries the main discriminative signal. Dense sentence embeddings (all-MiniLM-L6-v2) reached QWK = 0.5450, slightly above TF-IDF unigrams and bigrams (paired ΔQWK for E4 = −0.0110, 95% CI [−0.0199, −0.0027]). The two encodings therefore carry similar information; we keep TF-IDF in E4 because it is cheaper and its features are readable.

B. *Ordinal versus Multiclass Learning*

Training a true multiclass XGBoost model (E5mc, multi:softprob) on identical features and split gives QWK = 0.4736 with argmax decoding, so the ordinal model (E4, QWK = 0.5340) gains ΔQWK = +0.0604 (95% paired bootstrap CI [0.0495, 0.0718]). Argmax multiclass attains a higher raw accuracy (62.3% vs. 61.6% for rounded ordinal scores, or 54.9% under the optimized thresholds) and more predictions within one level (96.3% vs. 95.2%), but it collapses predictions onto the majority ESI-3 class (50.9% prevalence) and recovers only 9.7% of ESI-1 and 27.6% of ESI-2 visits (E4: 21.5% and 50.9%). The gap is a decoding effect: when the multiclass probabilities are reduced to an expected-class score and cut with ordinal thresholds, QWK is 0.5372, and the paired difference to E4 is −0.0032 (95% CI [−0.0110, 0.0041]). Ordinal regression also outperformed the neural ordinal baselines CORAL (QWK = 0.4139) and CORN (QWK = 0.4904) on these tabular features.

C. *Leave-One-Group-Out Feature Ablation*

Backward ablation removed one feature group at a time from the deployable model E4 (Table IV). Removing the chief complaint text caused the largest drop (ΔQWK = −0.0604, to 0.4736), so patient-reported symptoms are the most informative group, followed by administrative arrival features (0.0202). Engineered clinical scores (0.0072), raw vital signs (0.0058), demographics (0.0034), and comorbidities (0.0029) each changed QWK by less than 0.01, probably because the engineered composites overlap with the raw vitals and demographics.

TABLE IV. LEAVE-ONE-GROUP-OUT FEATURE ABLATION RESULTS (REFERENCE MODEL E4 = 0.5340)

Removed Feature Group	Remaining Dims	Resulting QWK	Marginal Impact (ΔQWK)
None (Full Reference Model E4)	640	0.5340	Reference (0.0000)
Chief Complaint Text (TF-IDF + Flags)	114	0.4736	−0.0604
Administrative / Arrival	625	0.5138	−0.0202
Engineered Clinical Scores	596	0.5268	−0.0072
Raw Physiological Vital Signs	630	0.5282	−0.0058
Patient Demographics	616	0.5306	−0.0034
Chronic Comorbidities	619	0.5311	−0.0029

D. *Detailed Minority Class and Error Distribution Analysis*

Table V and Fig. 2 give per-class metrics for the deployable model E4. Performance was highest for the majority classes: ESI-3 reached F1 = 0.612 (recall 55.8%, precision 67.7%) and ESI-4 reached F1 = 0.575 (recall 60.4%, precision 54.9%).

TABLE V. PER-CLASS DISCRIMINATION AND ERROR PROFILE FOR E4 (N_{TEST} = 11,625)

Acuity Class	Precision	Recall	F1	Support
ESI-1 (Resuscitation)	21.8%	21.5%	0.217	144
ESI-2 (Emergent)	37.8%	50.9%	0.434	1,694
ESI-3 (Urgent)	67.7%	55.8%	0.612	5,966
ESI-4 (Less Urgent)	54.9%	60.4%	0.575	3,345
ESI-5 (Non-Urgent)	22.2%	27.9%	0.247	476
Macro Average	**40.9%**	**43.3%**	**0.417**	**11,625**
E4: MAE = 0.5045 | Under = 22.2% | Over = 22.9% | Within ±1 = 95.2%				

For ESI-1 (1.5% prevalence, n = 144 test cases), E4 reached a recall of 21.5% and a precision of 21.8% (F1 = 0.217). As Fig. 2 shows, the largest share of the 113 missed ESI-1 cases (51, 45%) was classified as ESI-2 (emergent), which still alerts staff to serious illness; only 2 patients (1.4% of all ESI-1 cases) were assigned to non-urgent ESI-5. For ESI-5 (4.1% prevalence), recall was 27.9% (F1 = 0.247), and errors fell mostly in the adjacent ESI-4. For E4, undertriage was 22.2% and overtriage 22.9% (45.1% combined), slightly above the 12% to 45% mistriage range reported for nurse-assigned ESI levels [5].

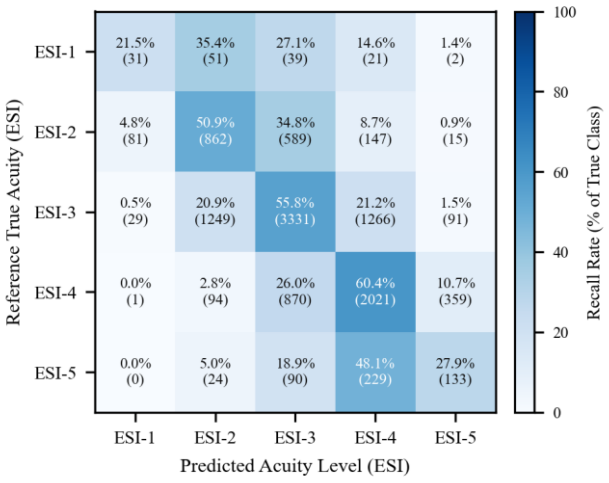


Fig. 2. Confusion matrix for primary deployable model E4 (row percentages denote clinical recall).

E. Model Explainability and Interpretability

We computed global feature importance (gain) and exact TreeSHAP values [24] for E4 on the whole test set. By mean absolute SHAP value, the strongest predictors were an indicator for vitals recorded during the visit (0.125), the age–comorbidity interaction (0.060), ambulance arrival (0.059), the complaint symptom density (0.044), and a high-risk complaint keyword flag (0.030); gain put the same first four features at the top, with ambulance arrival second. Higher values of the age–comorbidity interaction, the symptom density, the high-risk keyword flag, and ambulance arrival lowered the predicted score, that is, raised the predicted acuity.

F. Demographic Fairness and Wait-Time Gradient Audit

Cramér's V effect sizes between demographic subgroups and triage assignments (Fig. 3a) showed moderate associations for patient age (V = 0.152) and EMS arrival (V = 0.150) and small ones for insurance payer (V = 0.099), sex (V = 0.064), and race/ethnicity (V = 0.046). Chi-square tests reached $p < 0.001$ because of the large sample (N = 58,124), so we rely on the effect sizes: they indicate weak associations between these attributes and the recorded triage assignments. Subgroup evaluation of model errors remains future work.

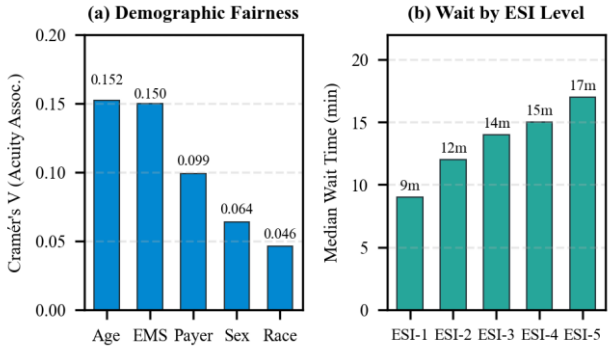


Fig. 3. Demographic fairness audit (a) and empirical wait-time acuity gradient (b).

Median wait times increased from ESI-1 to ESI-5 (Fig. 3b): 9 minutes for ESI-1, 12 for ESI-2, 14 for ESI-3, 15 for ESI-4, and 17 for ESI-5. An XGBoost model for wait time reached Spearman $\rho = 0.301$ (MAE = 28.7 min, $R^2 = 0.094$); the low R^2 reflects the absence of operational census data (e.g., bed availability, staffing) in NHAMCS. The raw data also show that Black patients waited a median of 16 minutes versus 13 minutes for White patients. This pre-existing disparity argues for active equity monitoring when AI decision-support tools are deployed.

V. Discussion

The findings suggest that structured machine learning can support ED triage when restricted to information available at presentation. We highlight three observations:

1) Decoding matters more than the loss: Cutting a risk score with QWK-optimized thresholds gains 0.0604 QWK over argmax multiclass classification (0.5340 vs. 0.4736) on identical features and split. When the multiclass probabilities are reduced to an expected-class score and cut with the same thresholds, QWK is 0.5372, so the gain comes from the decoding and not from the ordinal loss. Argmax multiclass keeps a higher exact accuracy (62.3%) and more predictions within one level (96.3% vs. 95.2%).

2) The chief complaint dominates: Removing the chief-complaint features causes the largest QWK drop (−0.0604), and TF-IDF unigrams and bigrams (0.5340) and dense sentence embeddings (0.5450) carry similar information, which suggests that explicit symptom terms carry clear acuity signals.

3) Assistive role: TRIAGEGEIST is intended as a second reader for triage nurses, not a replacement. It runs on standard CPU hardware and offers SHAP explanations that staff can inspect, but it needs prospective evaluation before any clinical use.

VI. Limitations and Future Work

The study has several limitations. All analyses use one US survey (NHAMCS), whose summarized encounter abstracts contain no real-time physiological streams, the analysis does not use the survey weights, and the model has not been tested on external multi-center EHR data. Missing values were median-imputed and the TF-IDF vocabulary was fitted on all rows, including the test set. ESI-1 is rare (1.5%), which limits sensitivity for resuscitation cases and keeps the overall QWK moderate (0.5340), and the reference labels carry inter-nurse subjectivity. Future work includes (i) external validation on multi-center EHR repositories such as MIMIC-IV-ED, (ii) domain-specific language models (ClinicalBERT, BioBERT) for nursing notes, (iii) cost-sensitive threshold tuning to raise ESI-1 sensitivity, and (iv) prospective silent-mode pilots of real-time decision support.

VII. Conclusion

We presented TRIAGEGEIST, an interpretable decision-support prototype for five-level ESI prediction. Treating ESI as an ordered scale and cutting a risk score with QWK-optimized thresholds on 58,124 NHAMCS visits preserves the order of acuity at modest computational cost. The deployable model (E4) uses only 640 triage-time features and reaches QWK = 0.5340 (95% CI: [0.5202, 0.5478]) with 95.2% of predictions within ±1 ESI level, ahead of argmax multiclass classification (0.4736) and the neural ordinal alternatives; multiclass probabilities decoded with the same thresholds perform on par, so the gain comes from the decoding. Ablation, SHAP explanations, a demographic audit, temporal proxy validation, and wait-time analysis support its use as a transparent, reproducible baseline for AI-assisted triage. External multi-center validation is the main next step. The code is available at https://github.com/cakho6179/triagegeist.

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